

# Security Linter Report

|                     |                          |
|---------------------|--------------------------|
| Filename            | case11_enterprise_app.py |
| Timestamp           | 2026-05-21 23:27:15 UTC  |
| Lines of Code (LOC) | 50                       |
| Verdict             | VULNERABLE               |

| ID         | Severity | Sink           | Source Variable | Source Type | Line |
|------------|----------|----------------|-----------------|-------------|------|
| CVE-SL-002 | WARNING  | cursor.execute | params          | EXTERNAL    | 71   |
| CVE-SL-003 | WARNING  | cursor.execute | params2         | EXTERNAL    | 72   |

|          |   |
|----------|---|
| CRITICAL | 0 |
| WARNING  | 2 |
| INFO     | 0 |
| TOTAL    | 2 |

**CVE-SL-002 — cursor.execute****WARNING**

|                              |                               |
|------------------------------|-------------------------------|
| <b>CWE</b>                   | CWE-89                        |
| <b>Sink</b>                  | cursor.execute                |
| <b>Sink Line</b>             | 71                            |
| <b>Sink Column</b>           | 0                             |
| <b>Sink Expression</b>       | cursor.execute(query, params) |
| <b>Argument</b>              | params                        |
| <b>Source Variable</b>       | params                        |
| <b>Source Type</b>           | EXTERNAL                      |
| <b>Propagation Mechanism</b> | data flow propagation         |

***Taint Path Trace:***

| Step | Role   | Line | Expression                             |
|------|--------|------|----------------------------------------|
| 1    | SOURCE | 41   | params = (col, table, col, safe_value) |
| 2    | SINK   | 71   | cursor.execute(query, params)          |

***Remediation:***

Identify the data source and use a parameterized query. Never concatenate external data into SQL strings:

```
# Replace string building with a parameterized query:  
cursor.execute("SELECT * FROM t WHERE col = %s", (external_value,))
```

**CVE-SL-003 — cursor.execute****WARNING**

|                              |                                 |
|------------------------------|---------------------------------|
| <b>CWE</b>                   | CWE-89                          |
| <b>Sink</b>                  | cursor.execute                  |
| <b>Sink Line</b>             | 72                              |
| <b>Sink Column</b>           | 0                               |
| <b>Sink Expression</b>       | cursor.execute(query2, params2) |
| <b>Argument</b>              | params2                         |
| <b>Source Variable</b>       | params2                         |
| <b>Source Type</b>           | EXTERNAL                        |
| <b>Propagation Mechanism</b> | data flow propagation           |

### ***Taint Path Trace:***

| Step | Role   | Line | Expression                      |
|------|--------|------|---------------------------------|
| 1    | SOURCE | 49   | params2 = (safe_category,)      |
| 2    | SINK   | 72   | cursor.execute(query2, params2) |

### ***Remediation:***

Identify the data source and use a parameterized query. Never concatenate external data into SQL strings:

```
# Replace string building with a parameterized query:  
cursor.execute("SELECT * FROM t WHERE col = %s", (external_value,))
```

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| Sink           | Line | Sanitizer | Protected Variable |
|----------------|------|-----------|--------------------|
| cursor.execute | 71   | int       | safe_new           |
| cursor.execute | 71   | int       | safe_cond          |

| Function | Variable  | Line |
|----------|-----------|------|
| int()    | safe_new  | N/A  |
| int()    | safe_cond | N/A  |

|                                          |                |
|------------------------------------------|----------------|
| CFG — Total Nodes                        | 45             |
| CFG — Condition/Loop Nodes               | 2              |
| CFG — Assignment Nodes                   | 23             |
| CFG — Call Nodes                         | 6              |
| CFG — Merge Nodes                        | 5              |
| ████████████████████████████████████████ | ██████████████ |
| DFG — Total Nodes                        | 86             |
| DFG — Total Edges                        | 90             |
| DFG — Source Nodes                       | 7              |
| DFG — Sink Nodes                         | 3              |
| DFG — Operator Nodes                     | 5              |
| DFG — Constant Nodes                     | 14             |
| DFG — Variable Nodes                     | 37             |
| Taint Propagation Hops                   | 33             |
| ████████████████████████████████████████ | ██████████████ |
| Analysis Time (ms)                       | 223.2          |

|                        |                                 |
|------------------------|---------------------------------|
| <b>Vulnerabilities</b> | 0 CRITICAL   2 WARNING   0 INFO |
| <b>Safe Sinks</b>      | 2                               |
| <b>Analysis Time</b>   | 223.2 ms                        |
| <b>Engine Version</b>  | 1.0.0                           |
| <b>Final Verdict</b>   | <b>VULNERABLE</b>               |